FLY LIKE AN EAGLE

BUILD A QUADCOPTER FROM SCRATCH

WHO IS THIS GUY?

I'M LOÏC

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SPIN42

THE IDEA

I WANTED TO FIND SOMETHING TO CHILL OUT

BIGGEST DRONE RACE

WORLD DRONE PRIX DUBAI

1st prize: $250,000

THE IDEA

THAT'S MY GAME!
LET'S BUY ONE


12A
AfroESC
12A
AfroESC

THE IDEA

I WANTED TO KNOW HOW IT WORKS.
LET'S BUILD ONE.

WHAT TO BUILD & FOR WHOM?

DON’T BUILD FOR /DEV/NULL

UNDERSTANDABLE

HUMAN FRIENDLY/ READABLE SOURCE CODE. WELL SEPARATED BUILDING BLOCKS

EXTENDABLE

EASY TO CONNECT HARDWARE, 3D PRINTED PIECES

DESIGN CHOICES

BUILDING BLOCKS

EVERYTHING TOGETHER

FURTHER WORK

DESIGN CHOICES

BUILDING BLOCKS

EVERYTHING TOGETHER

FURTHER WORK

Compute unit

Tools

Raspberry PI 3

Affordable

1GB memory

64bits Quad Core 1.2Ghz

17 GPIO (I2C, UART, ...)

Wifi

Compute unit

Tools

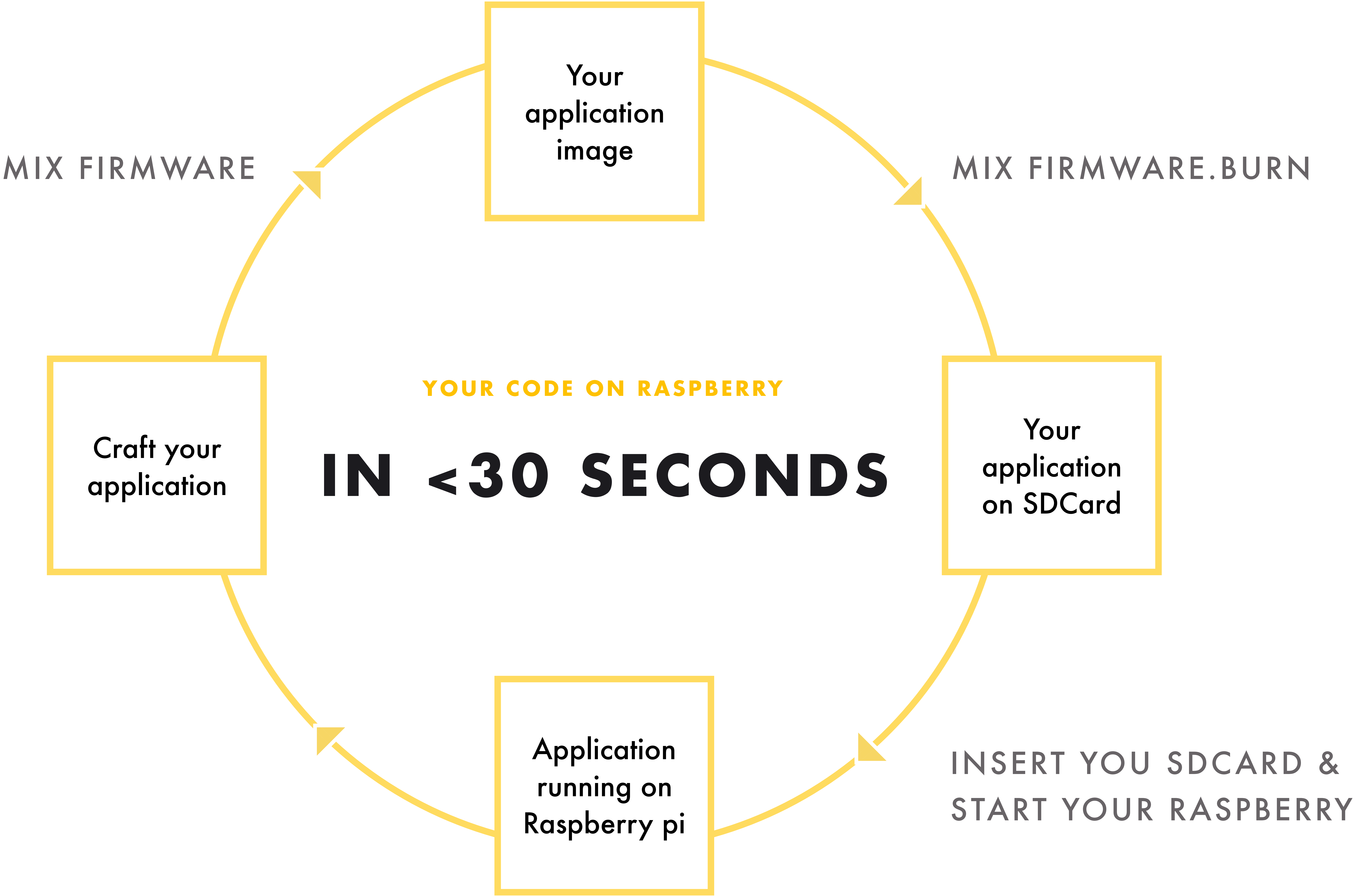
Nerves

Frameworks : Handle networks, I/O, ...

Tooling : Cross compiling tools for specific platform

Platforms : boot cross compiled linux directly to Erlang VM

DESIGN CHOICES



DESIGN CHOICES

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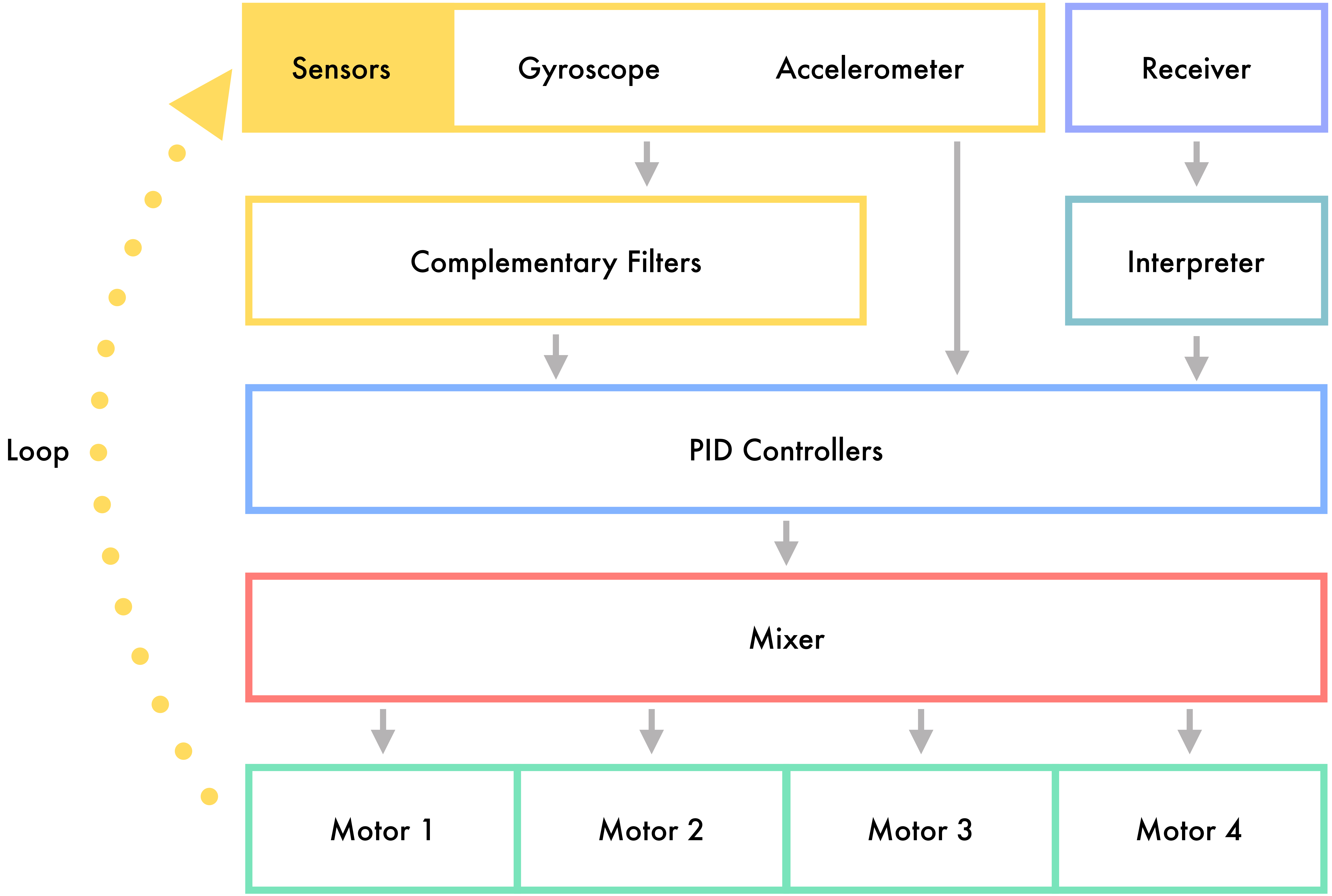
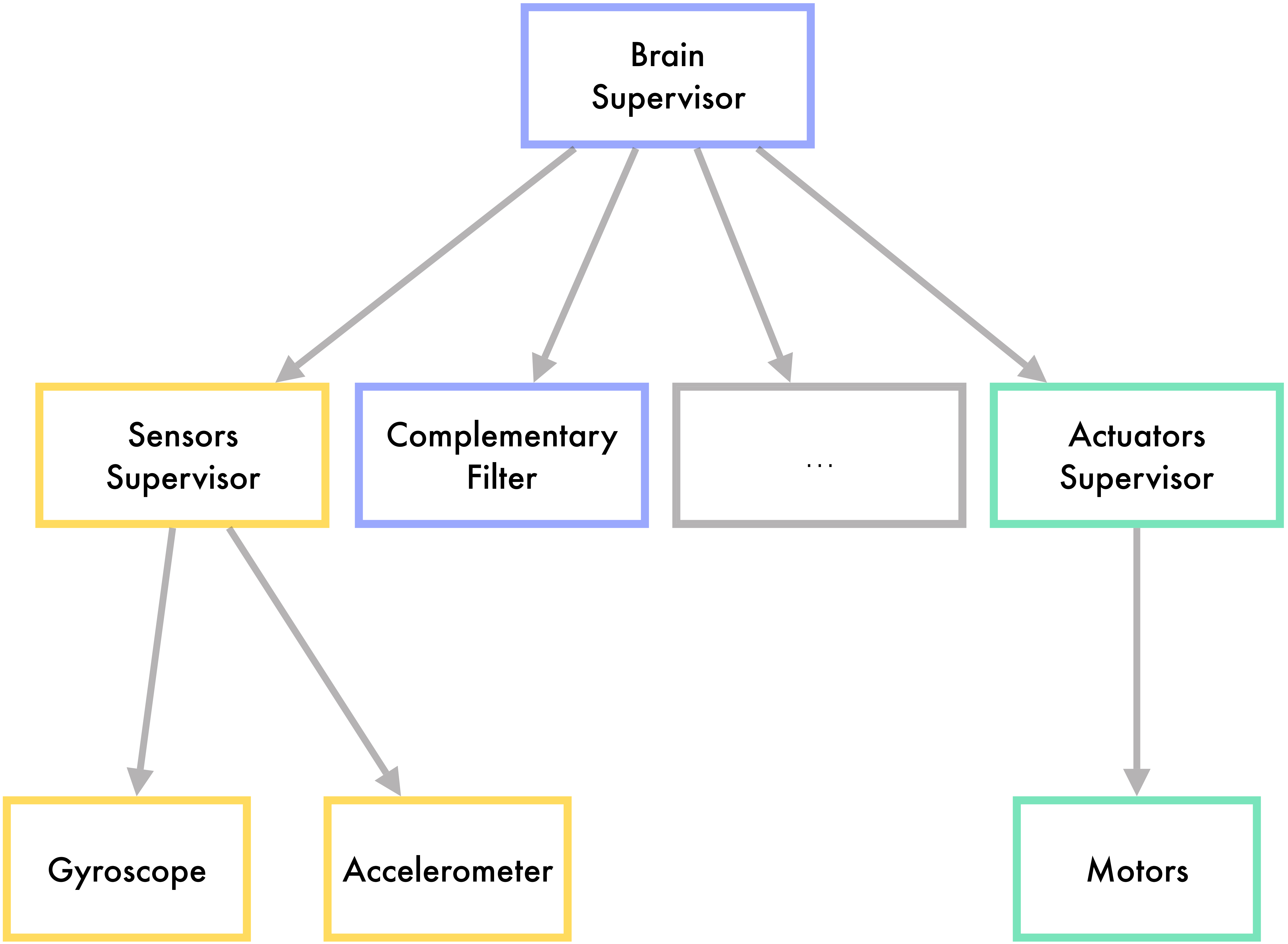


IT'S JUST A TREE

- Each component is a GenServer
- Components grouped under supervisors



```
config :brain, :sensors, [
  Brain.Sensors.Magnetometer,
  Brain.Sensors.Accelerometer,
  Brain.Sensors.Gyroscope
]
config :brain, Brain.Sensors.Gyroscope,
  driver: Drivers.L3GD20H
```

config.exs

sensor_supervisor.ex

```
defmodule Brain.Sensors.Supervisor do
  use Supervisor

  def init([]) do
    children = Application.get_env(:brain, :sensors) |> Enum.map(fn sensor_module ->
      sensor_configuration = Application.get_env(:brain, sensor_module)
      driver_module        = sensor_configuration[:driver]
      worker(sensor_module, [driver_module])
    end)
    supervise(children, strategy: :one_for_one)
  end
end
```

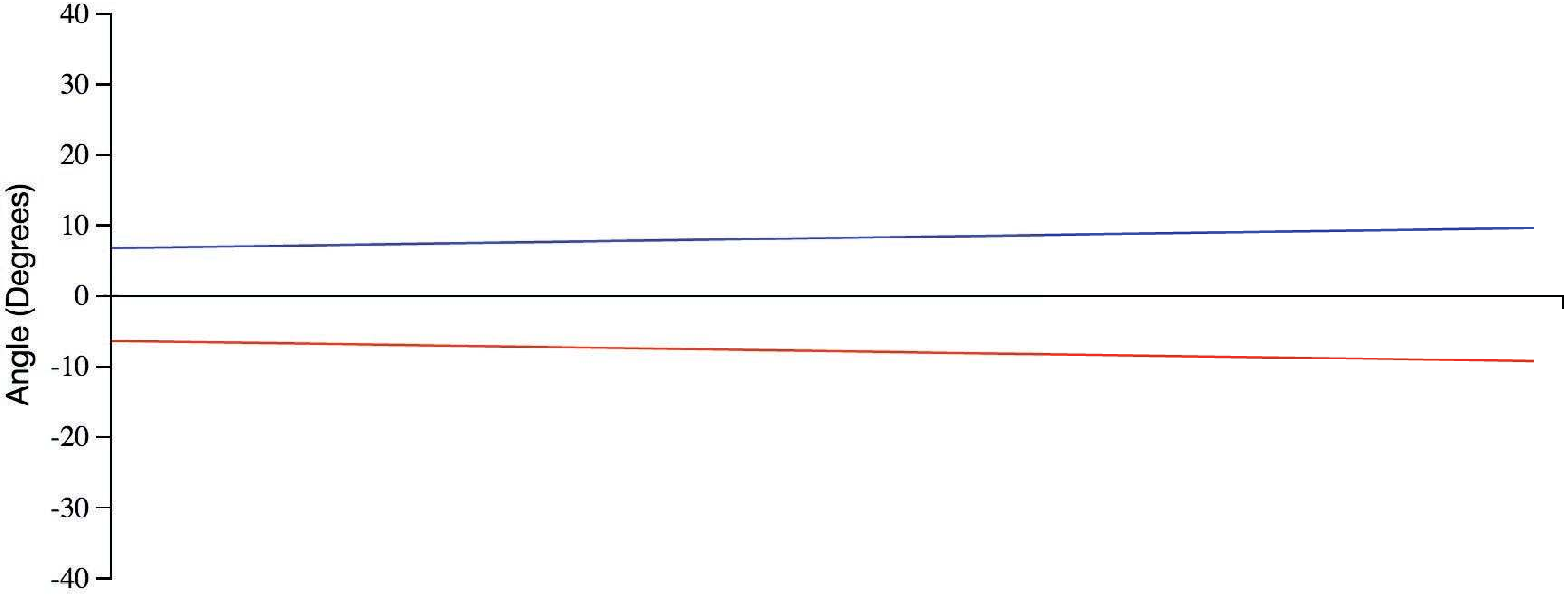
SENSORS

GET THE QUADCOPTER TILT

GYROSCOPE

Measures the angular speed (degrees/seconds)

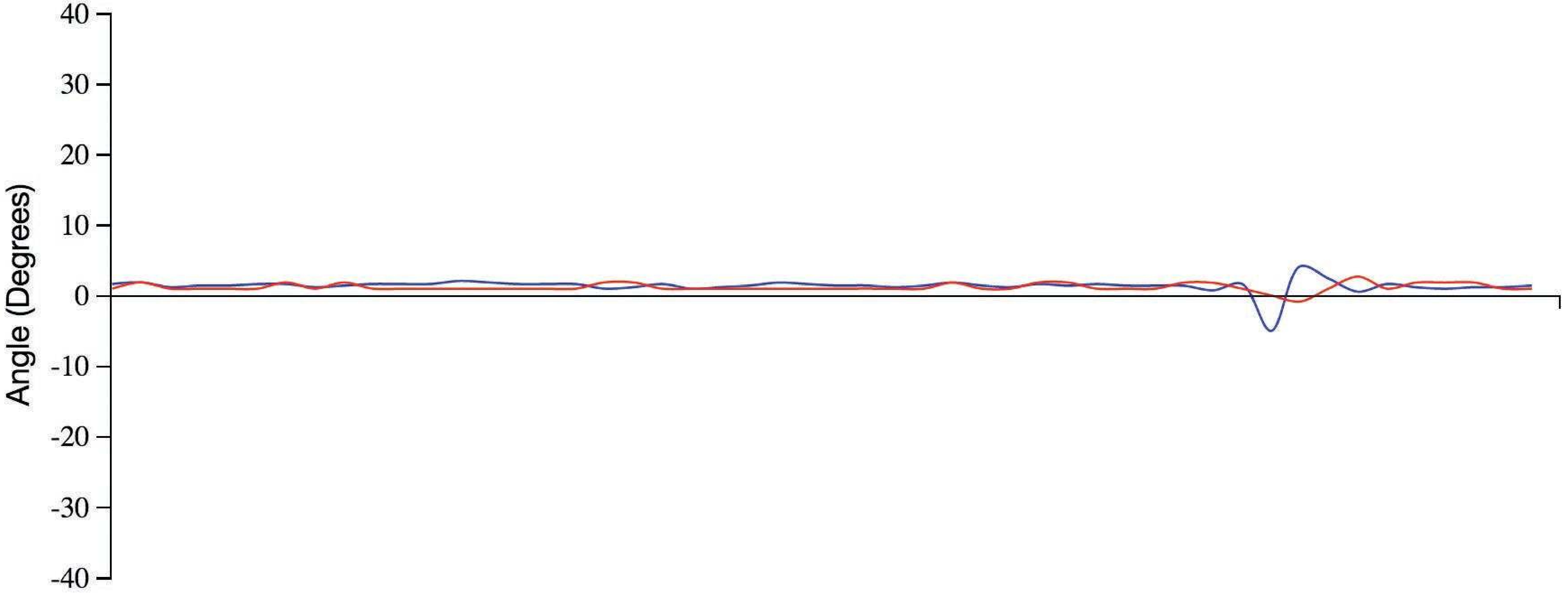
Its values drift with time



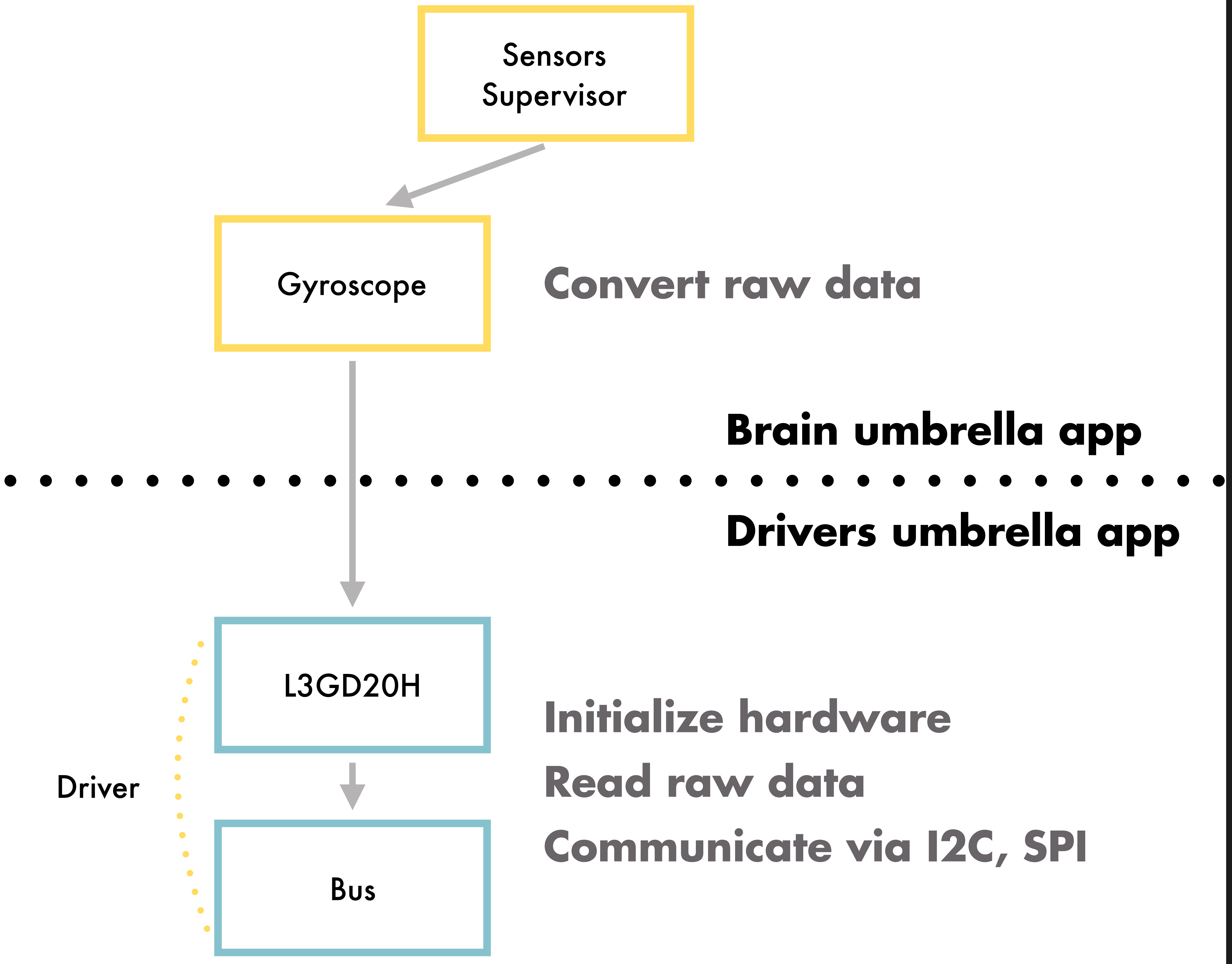
ACCELEROMETER

Measures the acceleration (g)

It is highly sensible to vibrations



BUILDING BLOCKS



IN PRACTICE

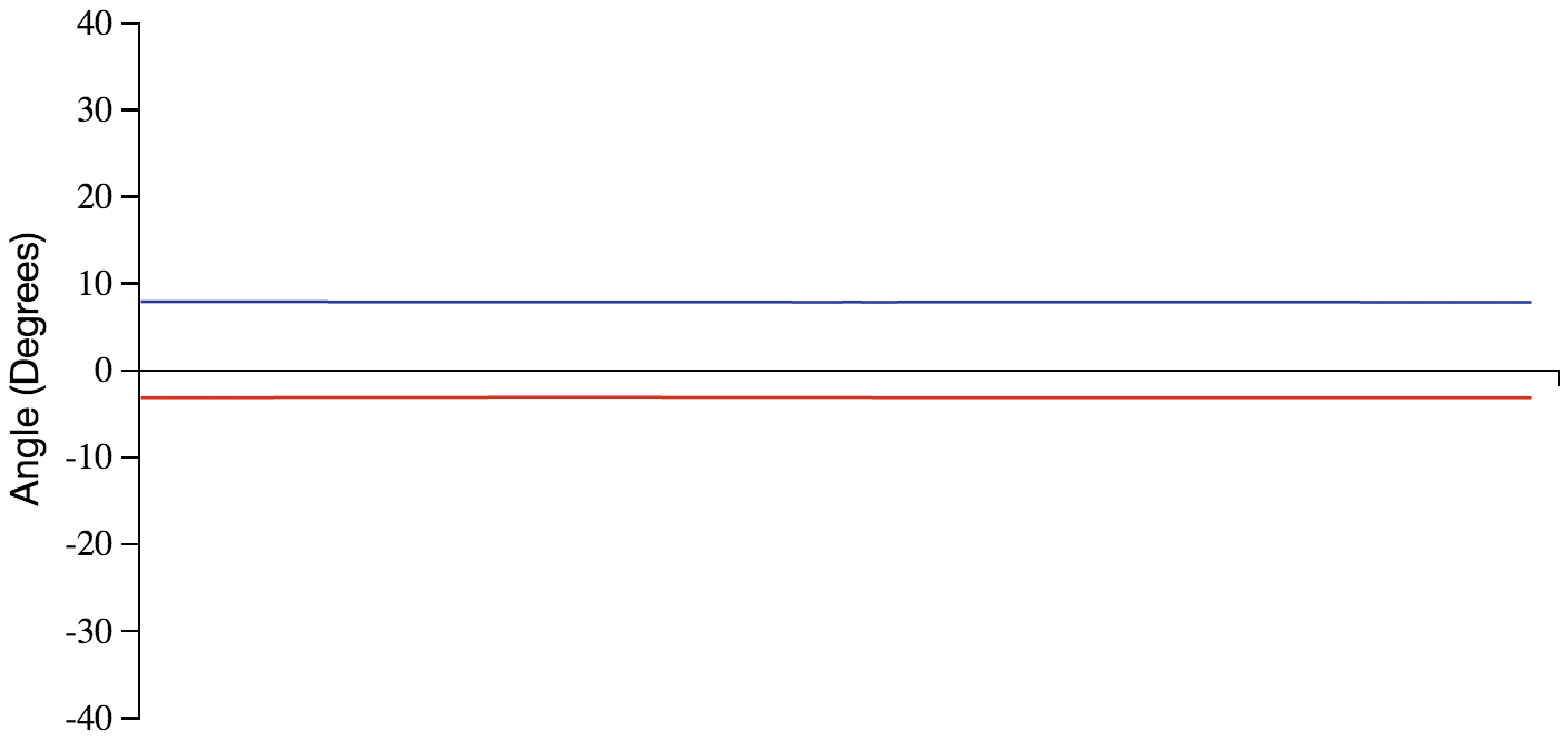
- Separate drivers and business logic
- Allow to switch driver for each sensor
- **elixir_ale** to handle communication

COMPLEMENTARY FILTER

Computes precise measure of the quadcopter tilt

Compensates accelerometer noise and gyroscope drift

angle = α * (gyroscope) + (1 - α) * accelerometer



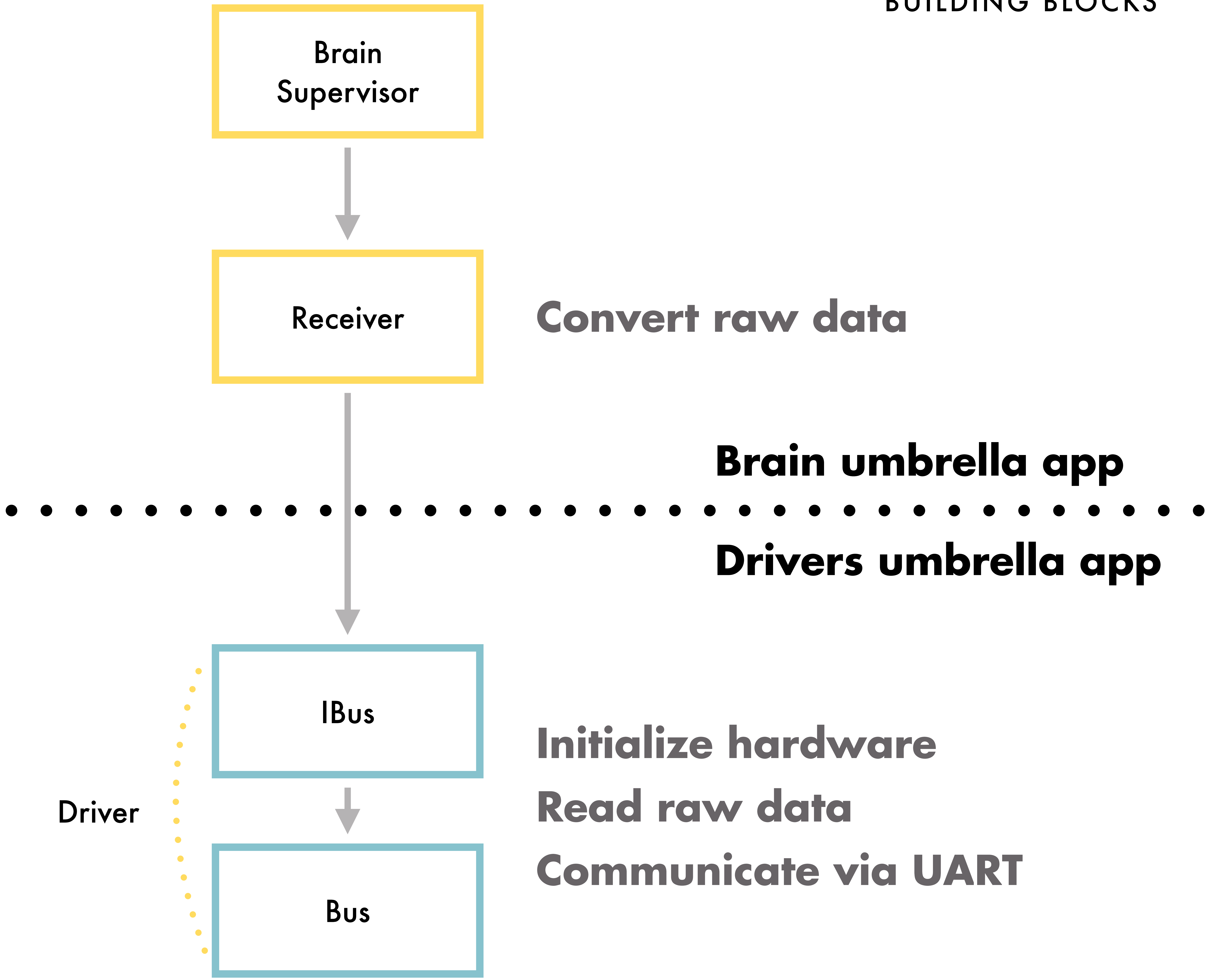
RECEIVER

RECEIVES COMMAND SENT BY USER TRANSMITTER

<<6, 0, 0, 1, 0, 0, 0, 255, 0, 0, 0, 2, 0, 0, 0, 255, 255>>

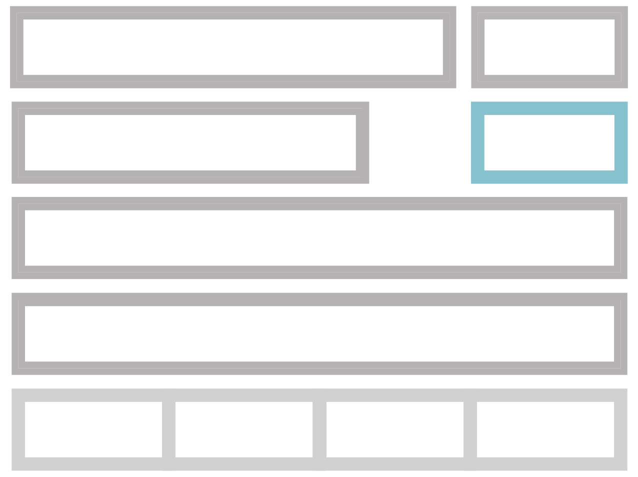
IN PRACTICE

- Same logic as sensors
- **nerves_uart** for serial communication

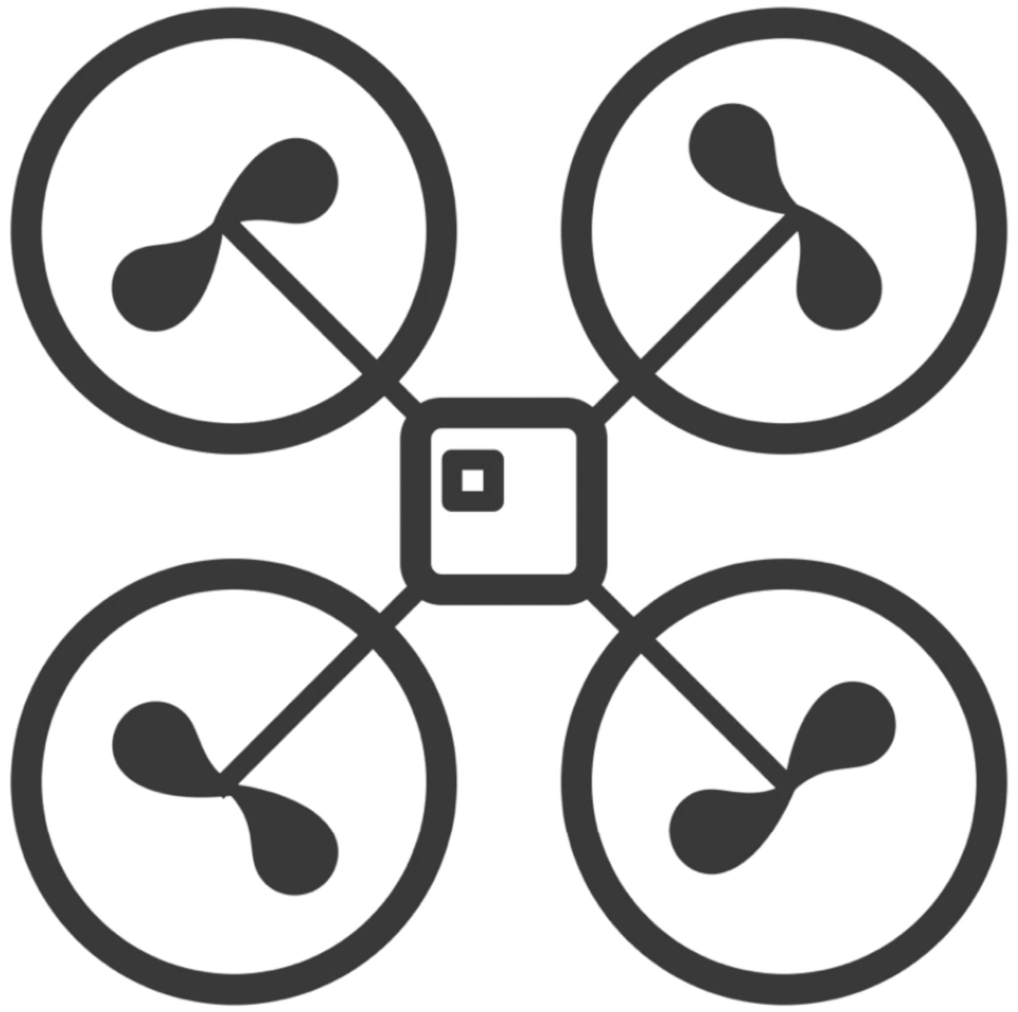


INTERPRETER

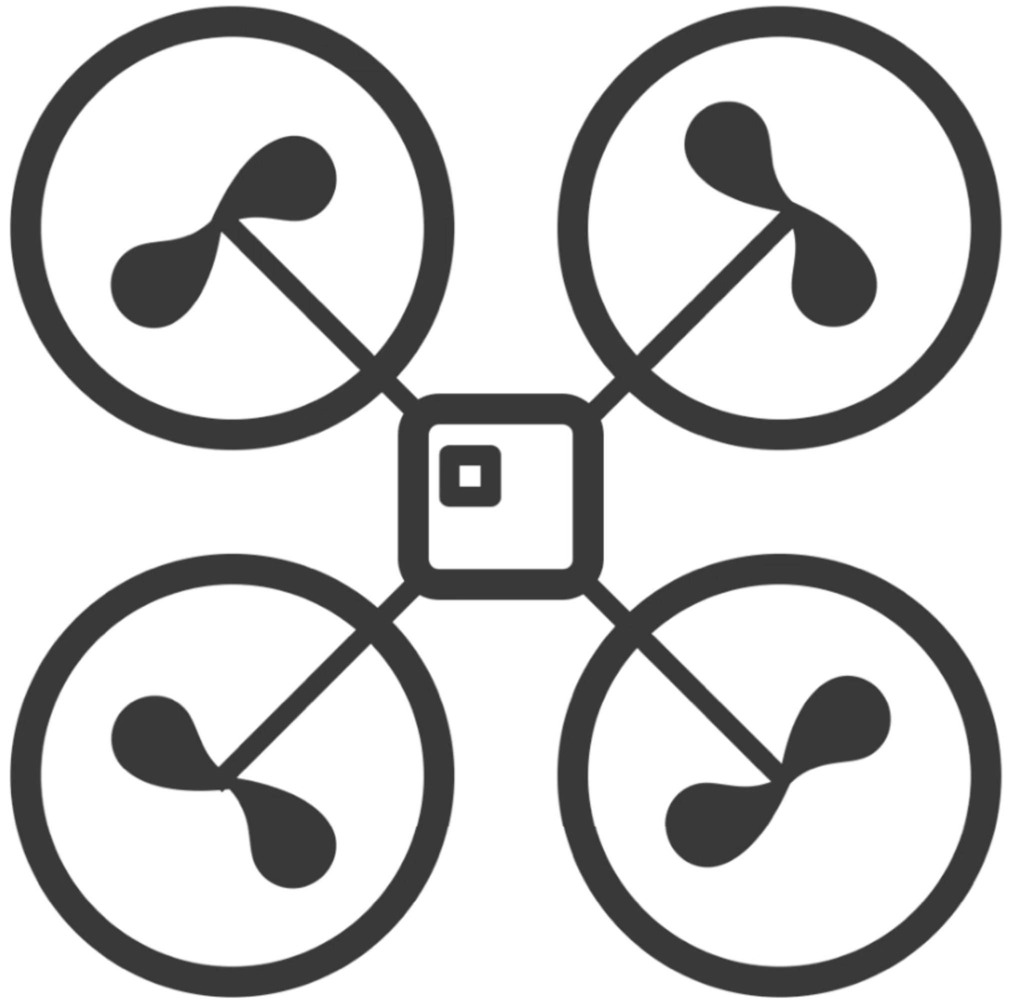
CONVERTS RECEIVER CHANNELS VALUES TO SET POINTS



RATE MODE

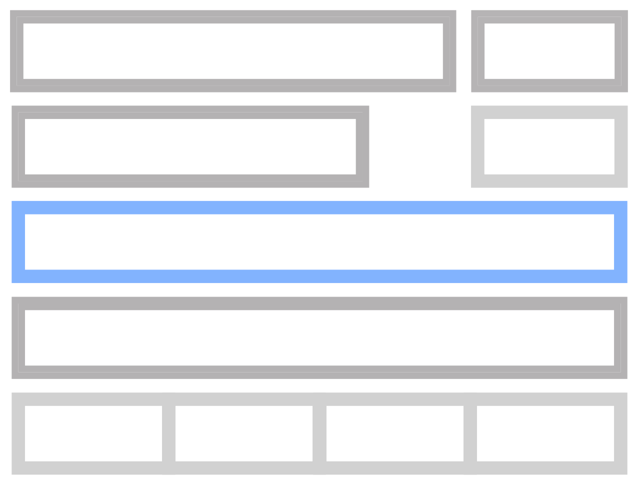


ANGLE MODE

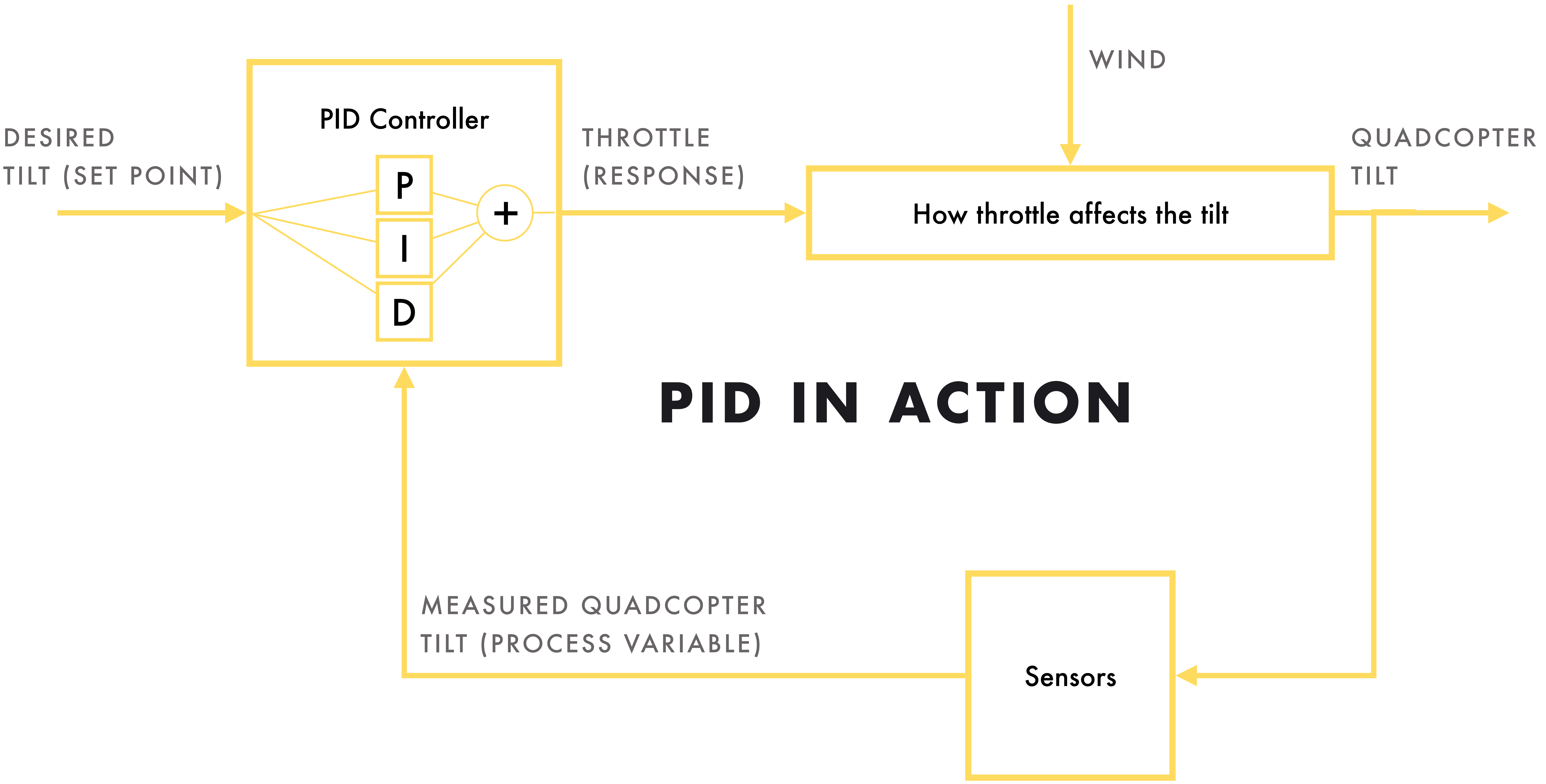


PID CONTROLLER

CONTROL LOOP FEEDBACK MECHANISM



PID IN ACTION



MIXER

COMPUTES THE SPEED TO APPLY TO EACH MOTOR

ROLL


BUILDING BLOCKS - MIXER
PITCH

YAW

DESIGN CHOICES

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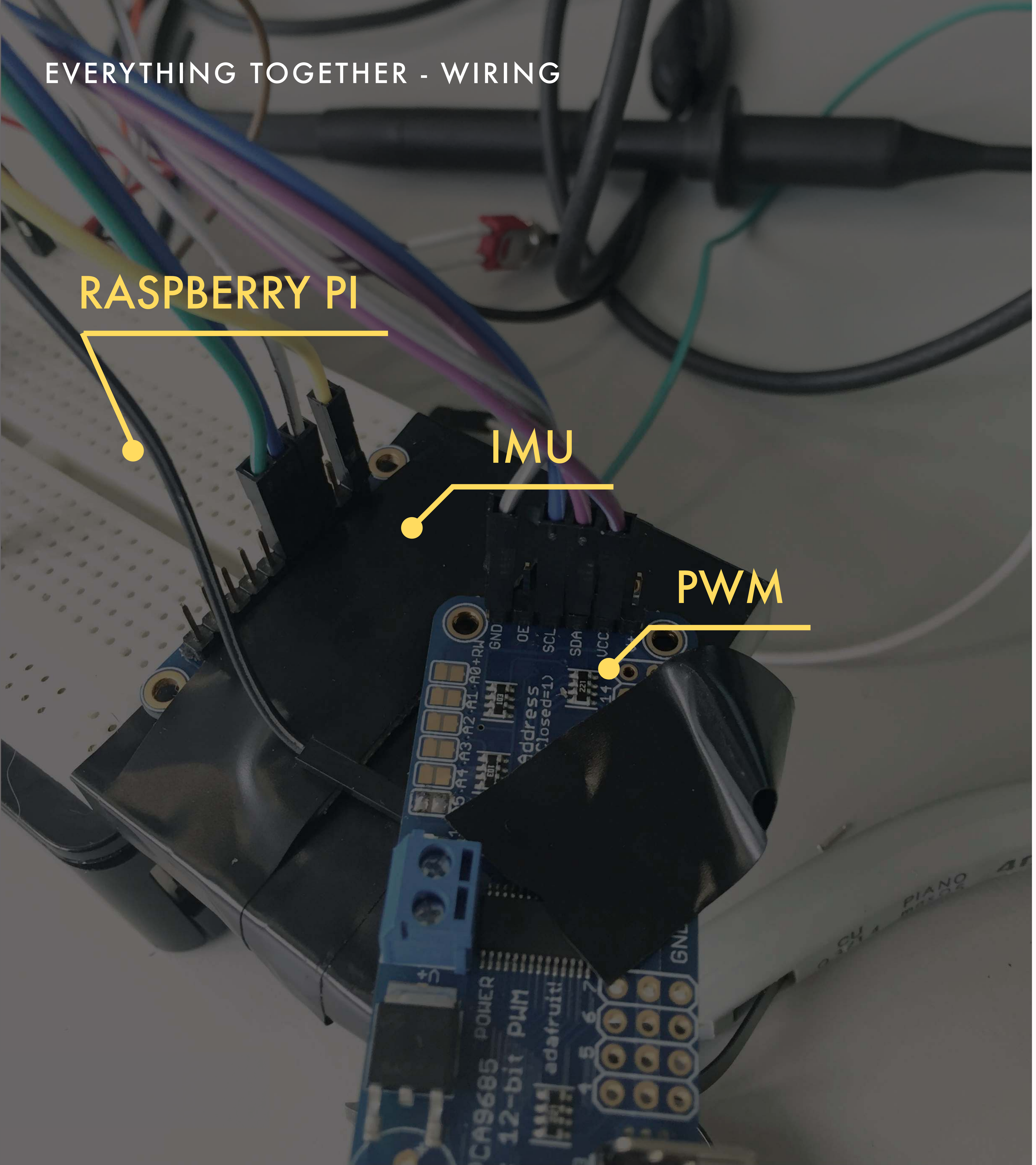
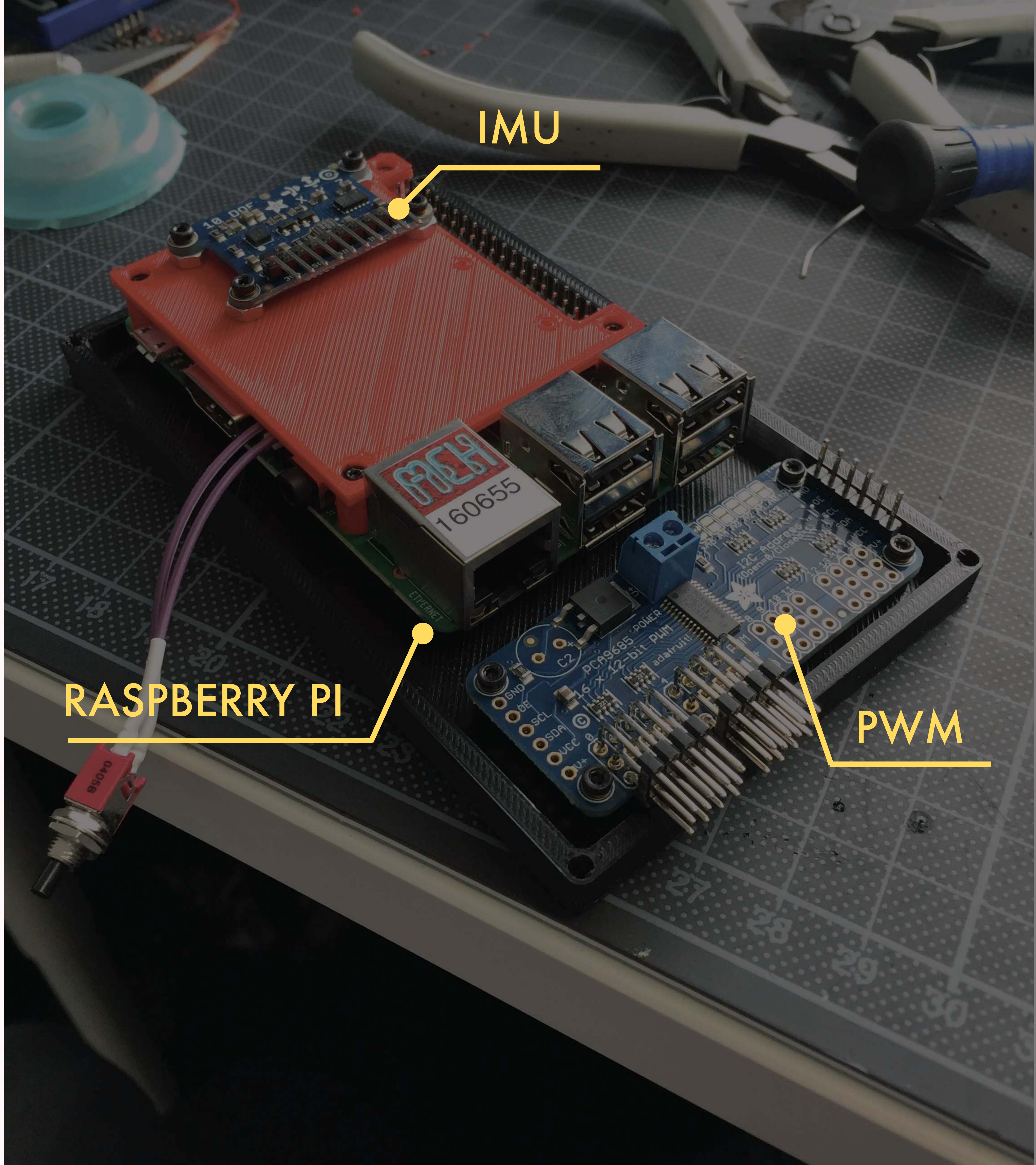
EVERYTHING TOGETHER

FURTHER WORK

EVERYTHING TOGETHER - WIRING





EVERYTHING TOGETHER - WIRING

EVERYTHING TOGETHER - WIRING


RECEIVER
ESC
RASPBERRY PI & SENSORS
MOTOR

OBVIOUSLY

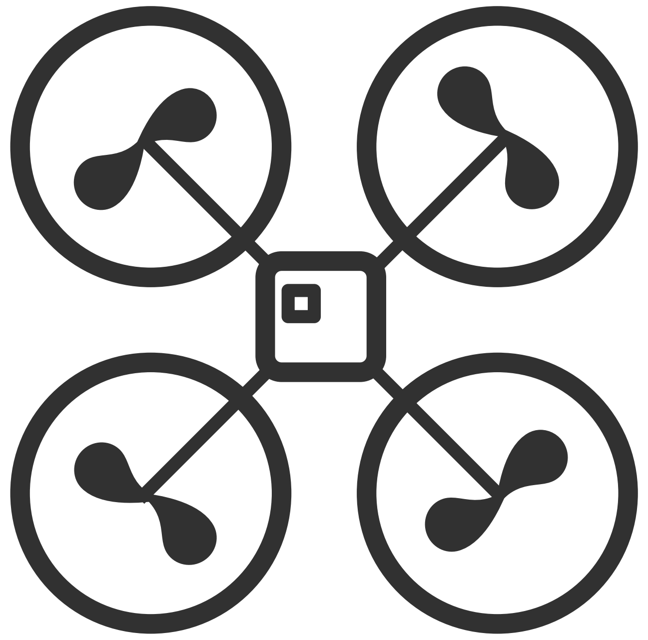
THAT'S A FAIL

DEBUGGING

REDUCE THE FEEDBACK LOOP

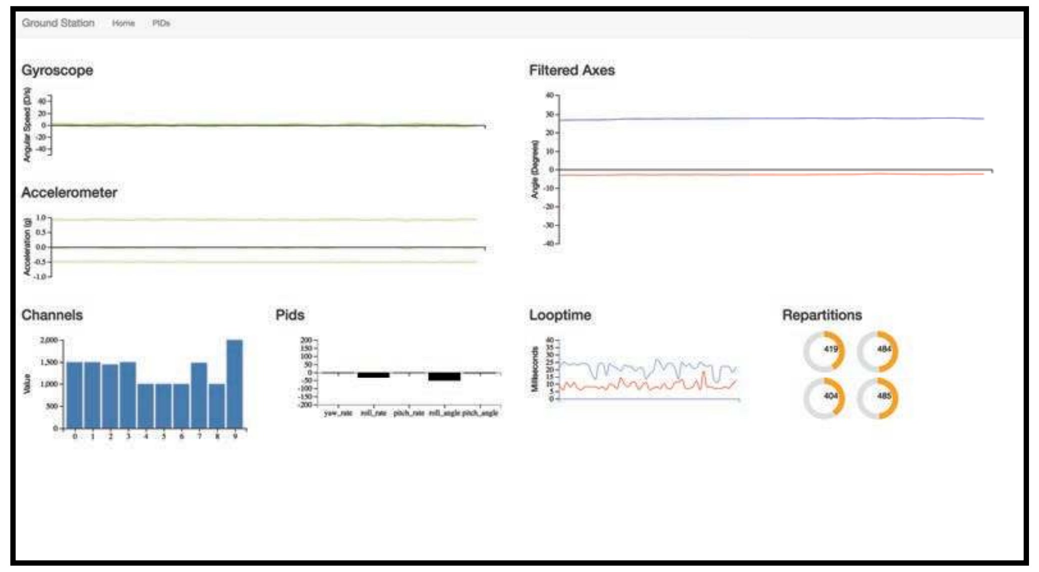
BLACKBOX

Collects data of each quadcopter components



Quadcopter

Phoenix Socket



Ground Station

Quadcopter Sensor

1

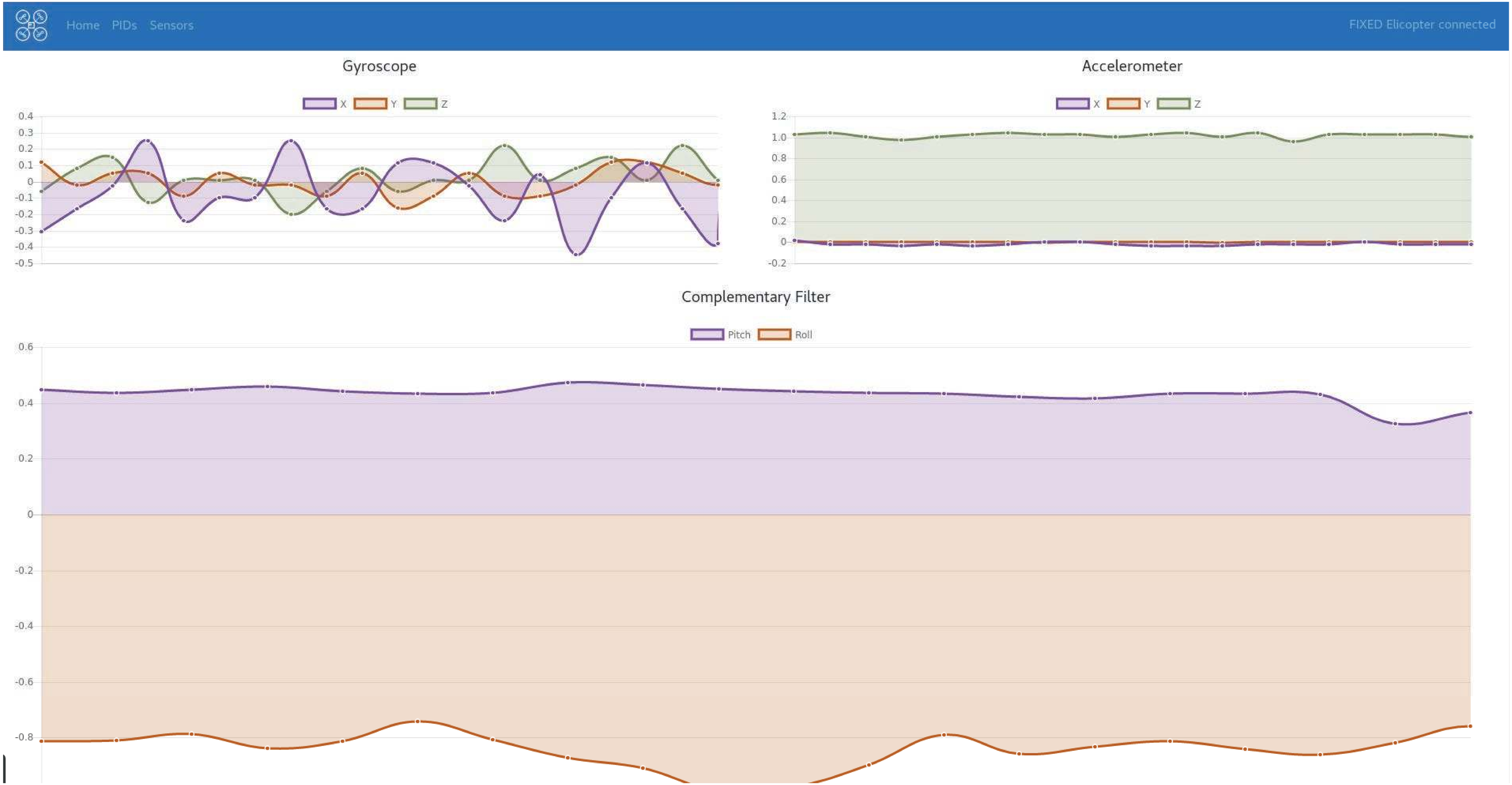
```
def handle_call(:read, _from, %{driver_pid: driver_pid} = state) do
  {:ok, data} = GenServer.call(driver_pid, :read)
  BlackBox.trace(__MODULE__, Process.info(self())[:registered_name], data)
  {:reply, {:ok, data}, state}
end
```

Blackbox

2

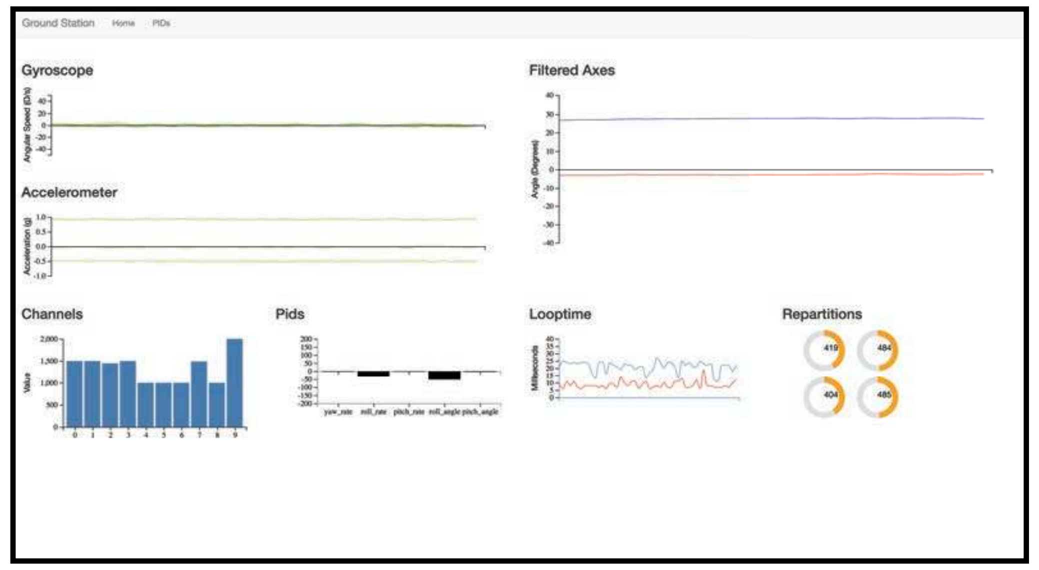
```
def handle_cast({:trace, key, data}, %{loops_buffer: loops_buffer} = state) do
  [last_loop | previous_loops] = loops_buffer
  last_loop                    = [{key, data} | last_loop]
  if length(previous_loops) >= @loops_buffer_limit do
    previous_loops = previous_loops |> Enum.drop(-1)
  end
  {:noreply, %{state | loops_buffer: [last_loop | previous_loops]}}
end

def trace(module, process_name, data) do
  event = case {module, process_name, data} do
    {Brain.Loop, _process_name, data} ->
      {:trace, :loop, data}
    {_, process_name, data} ->
      {:trace, process_name |> Module.split |> List.last |> Macro.underscore, data}
  end
  GenServer.cast(__MODULE__, event)
end
```

EVERYTHING TOGETHER - BLACKBOX



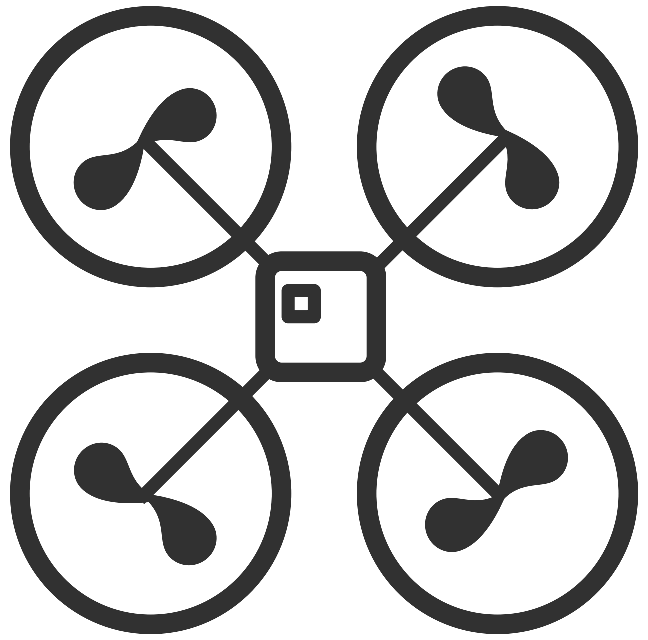
API

Tunes quadcopter in real time



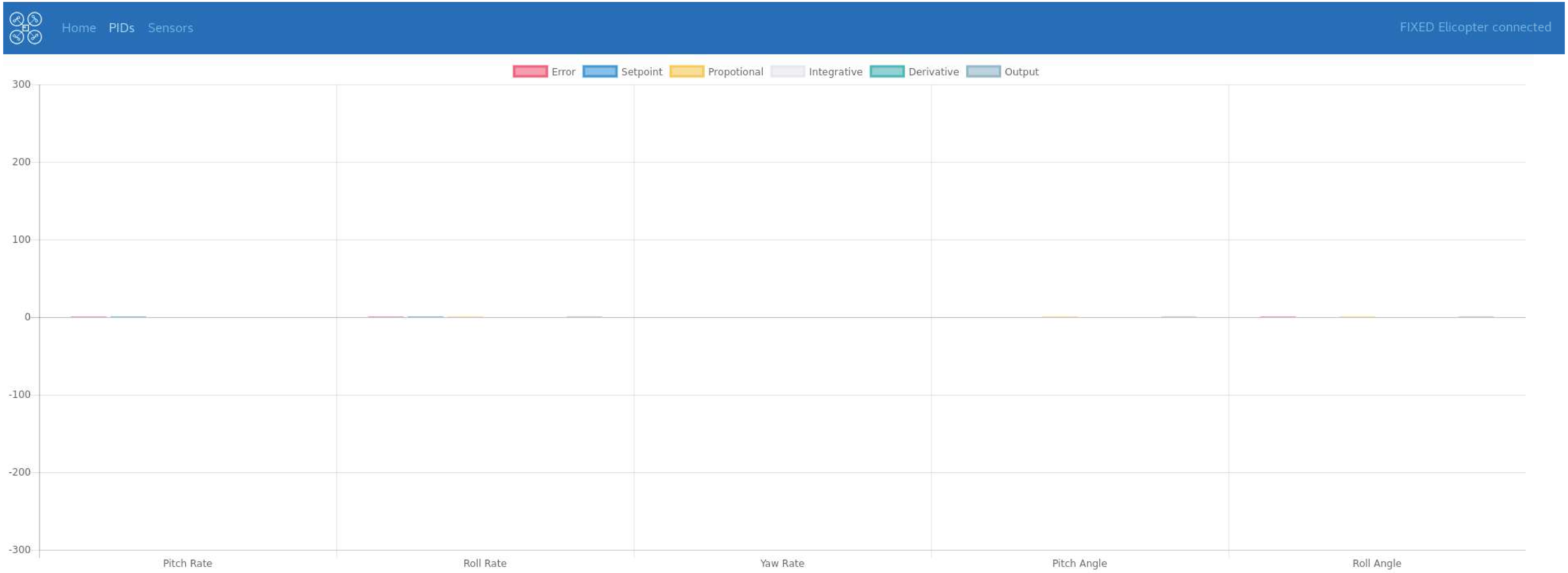
Ground Station

Phoenix REST API



Quadcopter

EVERYTHING TOGETHER - COMMANDER



Controllers

Name		kp			ki			kd		Minimum Output	Maximum Output
RollAnglePIDController	-	1.5	+	-	0	+	-	0	+	-400	400
PitchAnglePIDController	-	-1.5	+	-	0	+	-	0	+	-400	400
YawRatePIDController	-	1.7	+	-	0	+	-	0	+	-500	500
RollRatePIDController	-	0.7	+	-	0	+	-	0	+	-500	500
PitchRatePIDController	-	0.7	+	-	0	+	-	0	+	-500	500

SSDP DISCOVERY

- No need to change the IP address in the code
- **nerves_ssdp_server**
- **nerves_ssdp_client**

Quadcopter

1

```
def advertise_ssdp do
  name = Application.get_env(:brain, :name)
  {:ok, _name} = Nerves.SSDPServer.publish(name, "elicopter",[
    port: Application.get_env(:api, Api.Endpoint)[:http][:port]
  ])
end
```

Host

```
defp discover do
  Logger.info("Discovering Elicopters...")
  nodes       = Nerves.SSDPClient.discover
  elicopters = nodes |> Enum.filter(fn ({_name, info}) ->
    case info do
      %{host: _, st: "elicopter"} -> true
      _ -> false
    end
  end)
  {:ok, elicopters}
end
```

2

HTTP FIRMWARE UPDATE

- Use SSDP discovery
- No need to plug or remove the SD Card
- Auto reboot
- **nerves_firmware_http**

```
defmodule Mix.Tasks.Firmware.Upgrade do
  use Mix.Task
  require Logger
  def run(_) do
    {:ok, _pid}        = HTTPoison.start
    {:ok, _pid}        = Nerves.SSDPClient.start(nil, nil)
    {:ok, elicopters} = discover()
    {elicopter_name, elicopter_info} = elicopters |> List.first
     Logger.info("Found #{elicopter_name} at #{elicopter_info[:host]}.")
    build()
    upload(elicopter_info[:host])
    Logger.info("Firmware updated!")
  end

  defp discover do
    …

  defp build do
    …
  end

  defp upload(host) do
    Logger.info("Upload firmware...")
    firmware_update_url = "http://#{host}:8988/firmware"
    firmware_path       = Path.absname("_images/rpi3/brain.fw")
    HTTPoison.request!(
      :post, firmware_update_url, {:file, firmware_path},
      [{"Content-Type", "application/x-firmware"}, {"X-Reboot", "true"}],
      [timeout: 30_000, recv_timeout: 30_000]
    )
  end
end
```

1

2

3

PREPARE A SAFE DEBUGGING ENVIRONMENT

THE RESULT

AFTER SOME TUNING...


VIC
TO
RY!


DEPRECATED


Raspberry Pi 3 Model B V1.2
© Raspberry Pi 2015
GPIO
BROADCOM
ETHERNET

DESIGN CHOICES

BUILDING BLOCKS

EVERYTHING TOGETHER

FURTHER WORK

IMPROVE

INNOVATE

OPEN SOURCE PROJECT

DRAWINGS AND CODE ON GITHUB

GITHUB.COM/ELICOPTER

THANKS

GITHUB.COM/ELICOPTER